

西方经济学

Part 2 Classical Theory

THE ECONOMY IN THE LONG RUN

Lecture 2B A Monetary Model of a Closed Economy

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- (1) M11.2, M13.2; S4.4, S4. 附录, S6.2.¹
- (2) 其他资料: [观看](#)²电影《Die Fälscher (The Counterfeiters)》(2007), 电影《一出好戏 (The Island)》(2018)
- (3) 其他资料: [GM]
- A FIRST LOOK AT THE MONETARY SYSTEM
 - BANKING AND MONETARY POLICY
 - INFLATION: ITS CAUSES, EFFECTS, AND SOCIAL COSTS

¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

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- (1) 掌握古典经济学中的货币经济模型。
- (2) 理解古典二分法。
- (3) 领会马工程教材精神。



Outline

1 The Monetary System

- Creation of Money
- The Instruments of Monetary Policy
- Measuring Money

2 Classical Monetary Theory

- The Quantity Equation
- Cambridge Cash Balance Approach
- Determination of Price
- Inflation and The Fisher Effect
- Generalized Money Demand Function
- The Equilibrium in the Money Market
- Classical Dichotomy and Monetary Neutrality
- Horizons of Macroeconomics

3 马工程教材疑难重点



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Creation of Money

Private bank's balance sheet

Money is used for transactions and pays no interest. It includes two types of money: *Currency* (coins and bills) and *checkable deposits* (the bank deposits on which you can write checks). Bonds pay a positive interest but they cannot be used for transactions.

Activities	Assets		Liabilities	
Dwarf sells bonds to the central bank and deposits the money in private banks	Vault Cash (R)	$-b$	Checkable Deposits	$+b$
Loan to Tom	Vault Cash (R)	$+b(1-\theta)$		
	Loan	$-b(1-\theta)$		
Tom's deposits	Vault Cash (R)	$-b(1-\theta)$	Checkable Deposits	$+b(1-\theta)$
Loan to Jerry	Vault Cash (R)	$+b(1-\theta)^2$		
	Loan	$-b(1-\theta)^2$		
Jerry's deposits	Vault Cash (R)	$-b(1-\theta)^2$	Checkable Deposits	$+b(1-\theta)^2$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	Total Assets	$-b/\theta$	Total Liabilities	$+b/\theta$

The monetary base,¹ B , is defined as total liabilities of the central bank.

$$B = \text{Reserve deposits} + \text{Vault cash} + CU = R + CU.^2$$

$$M = D + CU.$$

$$\frac{M}{B} = \frac{D + CU}{R + CU} = \frac{1 + c_D}{\theta + c_D} \geq 1.$$

R denotes total bank reserves (Reserve deposits + Vault Cash);

CU denotes currency held by the nonbank public;

M denotes the money supply, the aggregate money, or the money stock;

D denotes checkable deposits;

θ denotes the banks' desired reserve-deposit ratio R/D ;³

c_D denotes the public's desired currency-deposit ratio CU/D .

¹ The monetary base is also called *high-powered money*, the *central bank money*, or *outside money*. Outside money is the quantity of money coming from outside of the private sector. [The New Palgrave](#).

² Reserve deposits or balances are balances held by depository institutions in master accounts and excess balance accounts at the central bank. Vault cash plus CU is called currency in circulation, which is outside the Treasury and the central bank.

³ A banking system with $\theta < 1$ is called fractional reserve banking while the one with $\theta = 1$ is called 100% reserve banking.

The Instruments of Monetary Policy

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1. How can the central bank change the monetary base?
 - a. Open-market operations.
 - b. Lending reserves to banks: discount window.
2. How can the central bank change the reserve-deposit ratio?
 - a. Reserve requirements.
 - b. Interest on reserves.



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Measuring Money

The monetary base, B , is also called M_0 .³ There are two widely used definitions of the overall money M^s (the aggregate money, the money supply, or the money stock):

- (1) $M_1 = \frac{1+c_D}{\theta+c_D} B$;
- (2) M_2 is composed of M_1 and other assets that are somewhat less moneylike but almost checkable.

The US money stock measures can be found at Federal Reserve Statistical Release (<http://www.federalreserve.gov/releases/H6/>).

³In China, M_0 is just measured as CU .

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The Quantity Equation or the Equation of Exchange

The Transactions Form of the Quantity Equation is formulated by Simon Newcomb (1885) and popularized by Irving Fisher (1911).

$$\text{Money} \times \text{Velocity} = \text{Price} \times \text{Transactions}, \quad M \times V = P \times T,$$

where V is the **transaction velocity of money** (the number of times money enters into transactions), T is the total number of transactions during some period of time, P is the price of a typical transaction.

Disadvantages: The number of transactions is difficult to measure. The volume of transactions includes final goods and services, intermediate goods, and existing assets.

The Income Form of the Quantity Equation: Pigou (1927) replaced gross transactions with income transactions in the quantity equation.

$$\text{Money} \times \text{Velocity} = \text{Price} \times \text{Output}, \quad M \times V = P \times Y,$$

where V is the **income velocity of money** (the number of times money enters income), Y is the output, and P is the price of one unit of output.

The quantity equation is an identity: the transaction form defines the velocity as $\frac{PT}{M}$ while the income form defines it as $\frac{PY}{M}$.

The role of money: It serves as the medium of exchange. It is 'in motion.'

From an Identity to a Theory

The Quantity Theory of Money

Assumption 1

The velocity of money is constant since it is determined by institutional factors and could be regarded as fixed for the short run.

Under assumption 1, the quantity equation (an identity) becomes the quantity theory of money:

$$M \times \bar{V} = P \times Y, \quad (1)$$

- (1) V is constant under assumption 1;
- (2) M is controlled by the central bank;
- (3) Y is determined by factor supply and the production function; and
- (4) P is determined endogenously.



Cambridge Cash Balance Approach

Monetary Market Interpretation

Pigou (1917), Marshall (1923), and Keynes (1923) assumed that the demand for money would be a proportion of income. The demand for *real money balances*, $\frac{M^D}{P}$, can be written as

$$\frac{M^D}{P} = kY,$$

where k is called the Cambridge constant, which is the real money balances people want to hold for every unit of income. The Cambridge interpretation of the quantity equation can be viewed as the equilibrium in the monetary market. That is, the demand for money, M^D , is equal to the supply of money, M , or equivalently,

$$\frac{M}{P} = kY, \quad k = \frac{1}{V}.$$

If people keep more money (a higher k), then money circulates more slowly (a lower V).

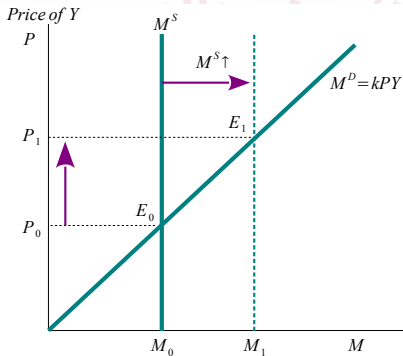
Advantages: It fits with the Marshallian demand–supply apparatus.

The role of money: It is a temporary abode of purchasing power. It is ‘at rest.’

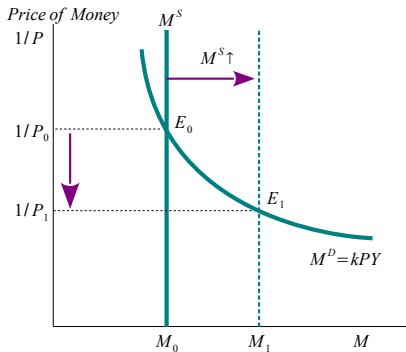


Determination of Price

The Equilibrium in the Money Market



(a) S(p.72)



(b)

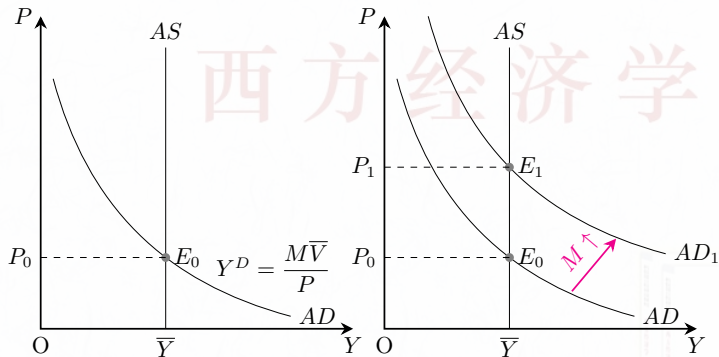
Nicolaus Copernicus (1517), *Memorandum on Monetary Policy*.



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Determination of Price

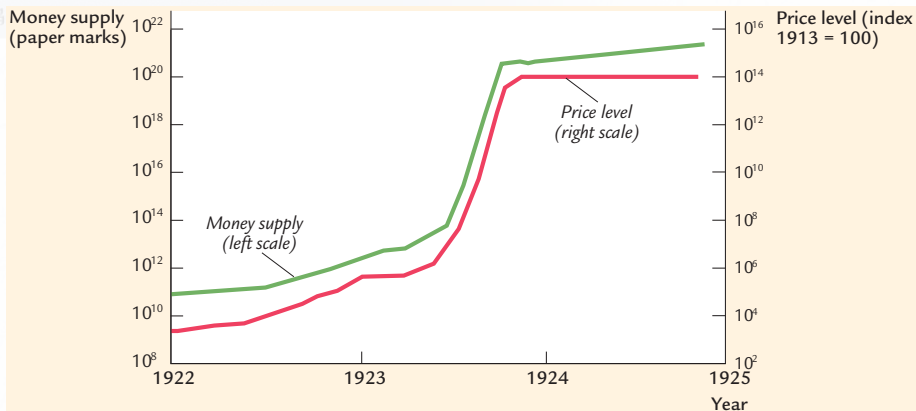
Schumpeter's AS-AD Framework of Classical Theory⁴



Example 1

中国人民银行：货币政策在总量上保持对实体经济的支持力度。…… 政策的重点是强调保持物价的稳定。

⁴ Schumpeter, Joseph A., 1954, History of Economic Analysis, Great Britain: Routledge.



The quantity of money and the general price level in Germany from January 1922 to December 1924.

Example 2

[观看](#) ^a电影 The Counterfeiters. 伯纳德行动如何影响盟国经济?

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The **Fisher equation** is given by

$$i_t = r_t + \pi_{t+1},$$

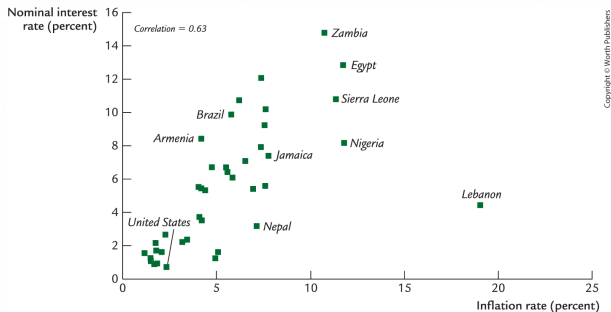
where $\pi_{t+1} = P_{t+1}/P_t - 1$ is the *ex post* inflation rate; r_t is the *ex post* real interest rate. The *ex ante* version of the Fisher equation is

$$i_t = r_t^A + \mathbb{E}_t \pi_{t+1},$$

where $\mathbb{E}_t \pi_{t+1}$ is the expected inflation rate; r_t^A is the *ex ante* real interest rate. In this case, r_t^A is determined by the equilibrium in the loanable funds market. An increase in $\mathbb{E}_t \pi_{t+1}$ by one percentage point leads to an increase in the nominal interest rate by one percentage point. The one-to-one relation between $\mathbb{E}_t \pi_{t+1}$ and the nominal interest rate is called the **Fisher effect**.



Evidence of the Fisher Effect



Inflation and Nominal Interest Rates Across Countries This scatterplot shows the average nominal interest rate on short-term Treasury bills and the average inflation rate in 40 countries from 2007 to 2022. The positive correlation between the inflation rate and the nominal interest rate is evidence of the Fisher effect.

Costs of Inflation

The costs of expected inflation:

1. Shoeleather cost of inflation. The Fisher effect implies that inflation raises the nominal interest rate which lowers the demand for real money balances. Thus people have to go to banks more frequently than before.
2. Menu costs. Producers have to change their posted prices more frequently than before.
3. Microeconomic inefficiencies due to variability of relative prices when producers cannot change their prices flexibly.
4. Altering tax burden, often in ways that lawmakers did not intend.
5. Inconvenience of living in a world with a changing standard. Money is a standard with which we measure economic transactions.

The costs of unexpected inflation:

1. Redistribution of wealth among individuals.
2. It hurts individuals living on fixed income.
3. Individuals with imperfect information often have price-misperceptions which distort their decisions.

One benefit of inflation: A 2 percent wage cut in a zero-inflation world is, in real terms, the same as a 3 percent raise with 5 percent inflation.

Generalized Money Demand Function

The Theory of Liquidity Preference

"The three divisions of liquidity-preference which we have distinguished above may be defined as depending on (i) the transactions-motive, i.e. the need of cash for the current transaction of personal and business exchanges; (ii) the precautionary-motive, i.e. the desire for security as to the future cash equivalent of a certain proportion of total resources; and (iii) the speculative-motive, i.e. the object of securing profit from knowing better than the market what the future will bring forth."

—Chapter 13.II, *The General Theory of Employment, Interest and Money*,
by JOHN MAYNARD KEYNES (1936)

Generalized money demand function can be written as

$$\frac{M^D}{P} = L\left(Y, i\right),$$

where Y denotes real income and i nominal interest rate.

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Generalized Money Demand Function

By the *ex ante* version of the Fisher equation,

$$\frac{M^D}{P} = L\left(Y, r + \pi^e\right),$$

where Y and r are determined by the real side of the classical economy while the expected inflation rate, π^e , is taken as given. The equilibrium in the monetary market is characterized by

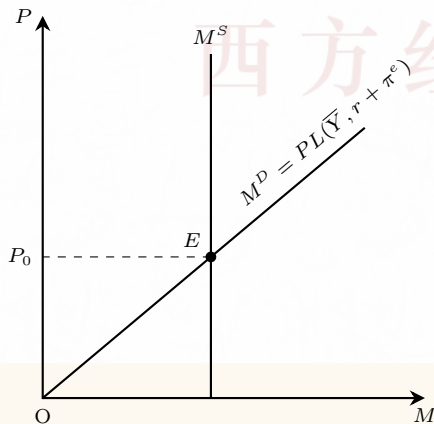
$$\frac{M}{P} = L\left(Y, r + \pi^e\right), \quad (\text{LM})$$

where the money supply, M , is controlled by the central bank. The above equation is referred to as the LM equation.

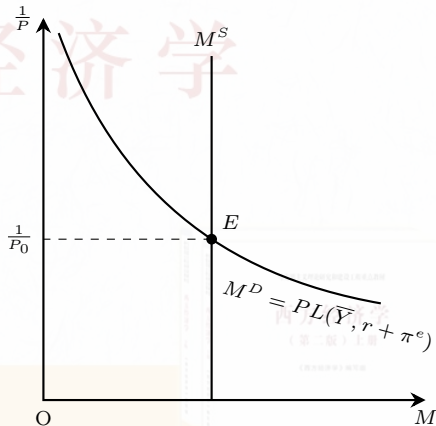


The Equilibrium in the Money Market

Monetary neutrality: Money is a veil



(a)



(b)



Classical Dichotomy and Monetary Neutrality

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Supply of goods :

$$Y = F(\bar{K}, \bar{N}; A);$$

Demand for goods :

$$Y = C(Y - T) + I(r) + G;$$

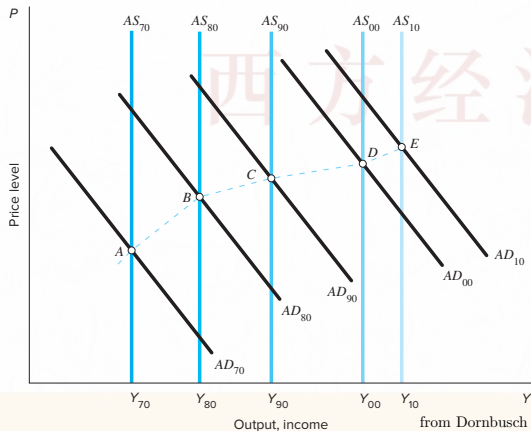
The LM equation :

$$\frac{M}{P} = L(Y, r + \pi^e).$$



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Horizons of Macroeconomics



Horizons of Macroeconomics:

- (1) In the short run, aggregate supplies of factors are fixed ($\bar{K}, \bar{N}$) while wages and prices are sticky. Involuntary unemployment of factors *could* occur ($N < \bar{N}, K < \bar{K}$).
- (2) In the long run (**the quasi-long-run**), aggregate supplies of factors are fixed ($\bar{K}, \bar{N}$) while wages and prices are flexible. There is no involuntary unemployment of factors ($N = \bar{N}, K = \bar{K}$).
- (3) In the very long run (**the long run**), aggregate supplies of factors change; wages and prices are flexible.



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疑难重点

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- (1) 货币是如何创造的？货币乘数如何计算？了解货币的度量口径。
- (2) 掌握 the quantity theory of money, Cambridge cash balance approach, the Fisher effect.
- (3) 预期到的和非预期到的通胀，会产生什么成本？
- (4) 古典经济学中，价格是如何决定的？什么是货币中性？什么是古典二分法？
- (5) 持有货币的三种动机是什么？



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1 (E2, p.176)

根据马工程教材观点，应当如何评析西方经济学的通货膨胀理论？

2 (E2, p.349)

根据马工程教材观点，资本主义经济为什么不能像一般均衡理论所描述的那样实现所有市场的供求均衡？

